

Day 2 – 2,3



# Weakly-supervised learning

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# Schedule

Each hour is scheduled as **50 min lecture + 10 min break**

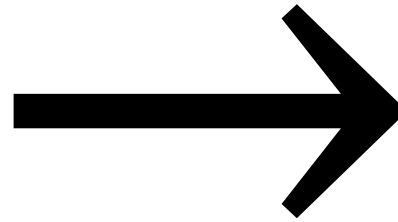
- **Day 1:** 4 hr lecture + 4 hr practice
  - **AM:** CNN, Transformers, Object detection
  - **PM:** DETR, MI-DETR
- **Day 2:** 4 hr lecture + 4 hr practice
  - **AM:** Self-supervised learning, Weakly supervised learning, Multimodal models, Semantic segmentation
  - **PM:** DINO, SAM variants

Two (competing) goals in machine learning

Better fitting the model to data

Generalization

Models try to map data to label



Chair

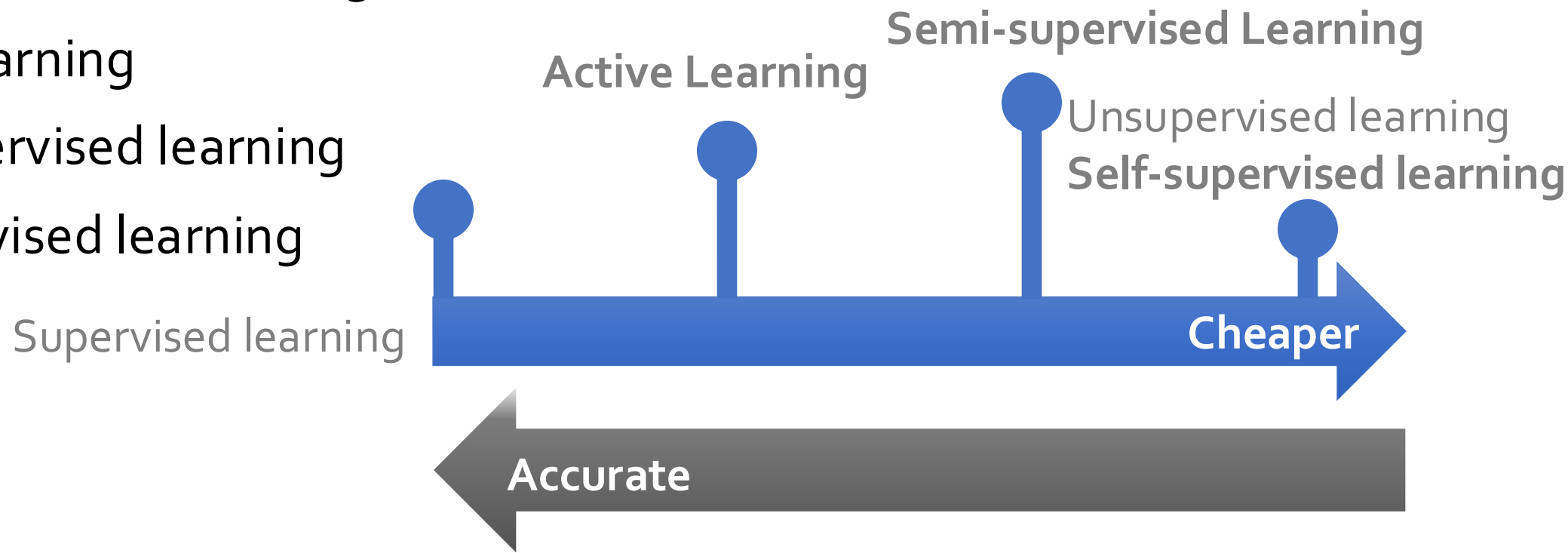
# What makes a chair a chair?



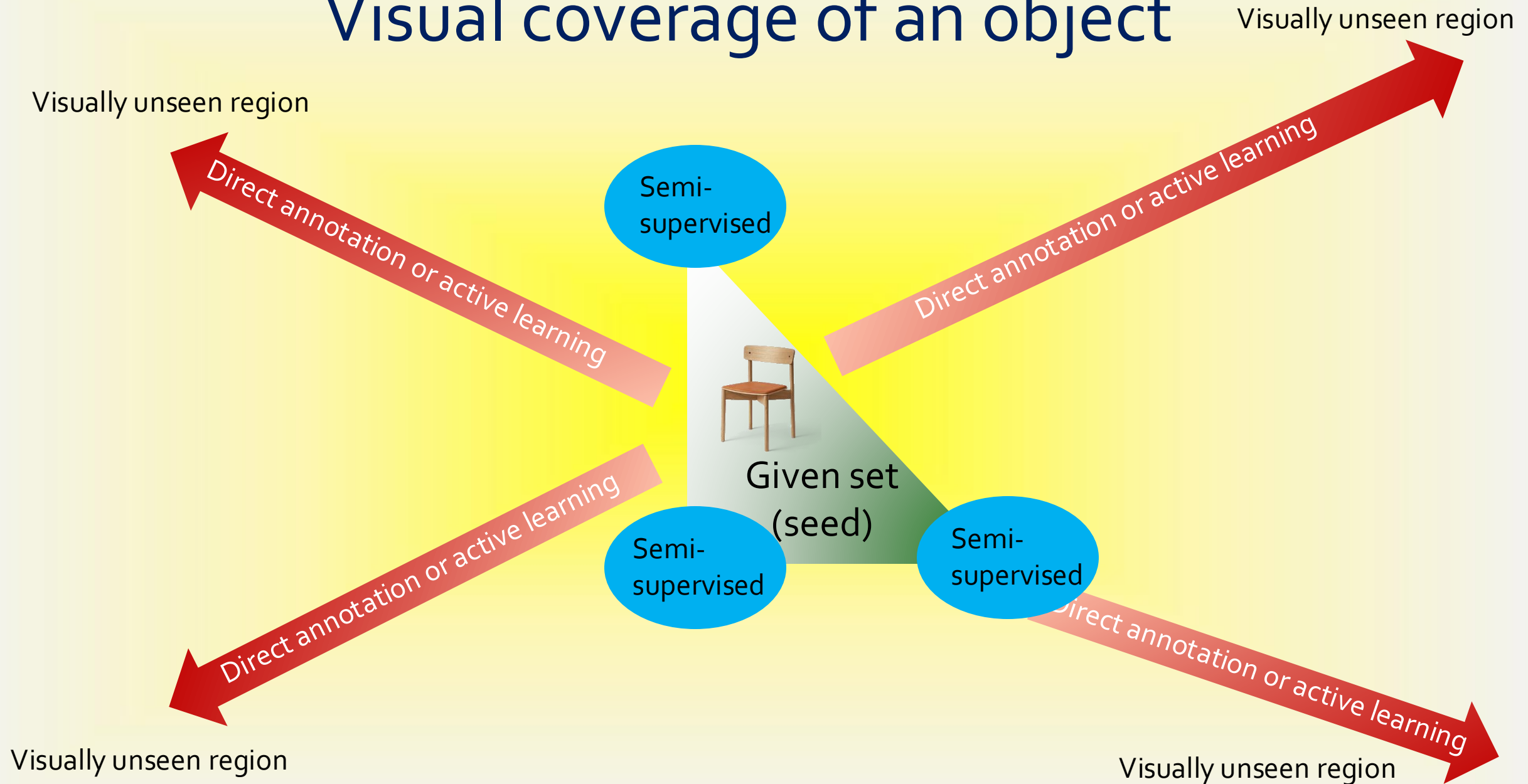
Visually diverse but shares same semantics (affordance or functionality)  
- *i.e.*, something to sit on

# Remedies in the literature

- Large scale supervised learning
- Few-shot learning
- Semi-supervised learning
- Active learning
- Self-supervised learning
- Unsupervised learning



# Visual coverage of an object



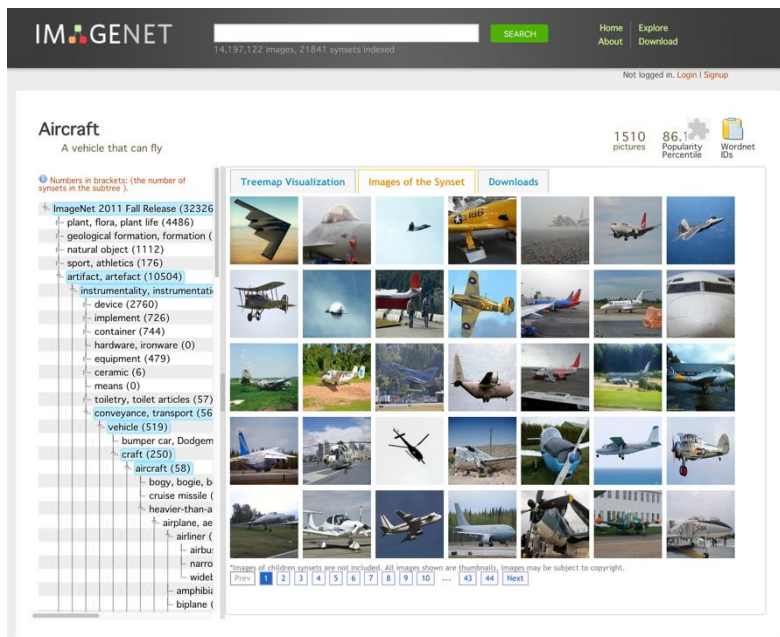
# Supervised Classification: find $f$

- **Given:** Training set  $\{(x_i, y_i) \mid i=1, \dots, n\}$
- **Find:** A good approximation to  $f: X \rightarrow Y$  (called **Hypothesis**)
- **Examples:** what are  $X$  and  $Y$ ?
  - Spam detection
    - Map email to  $\{\text{Spam}, \text{Ham}\}$
  - Digit recognition
    - Map pixels to  $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
  - Stock prediction
    - Map new, historic prices, etc. to the real numbers (money)!



# Large scale learning around 2012

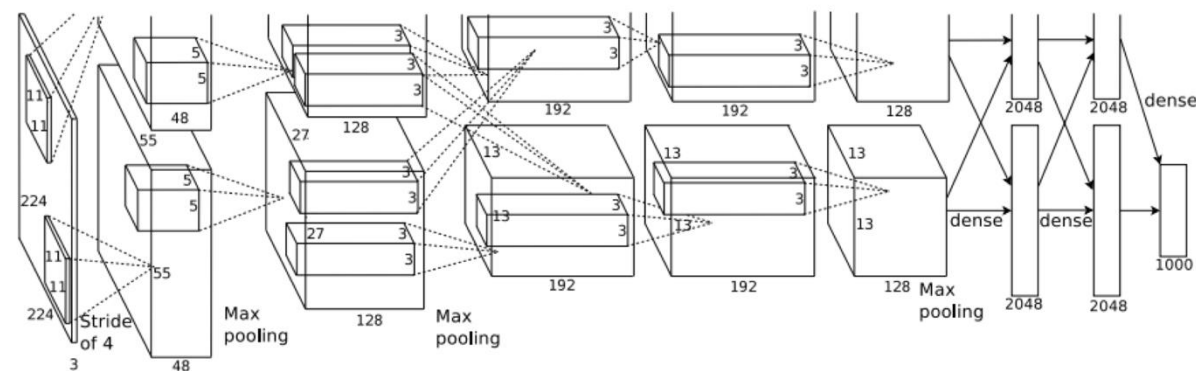
## Large labeled datasets



ImageNet, Deng et al., CVPR 2009

## Some deep neural net architecture

+



Krizhevsky et al., NeurIPS 2012



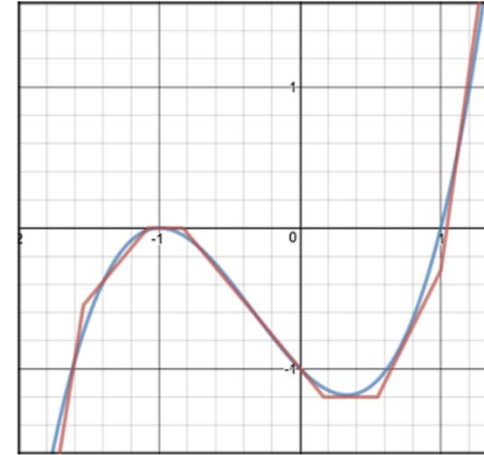
# Why high performance model is data hungry?





# Universal Approximation Theorem

- A feed-forward network with
  - a single hidden layer
  - containing a finite number of neurons
  - can **approximate continuous functions**
    - on compact subsets of  $\mathbf{R}^n$
    - under mild assumptions on the activation functions
- A simple neural networks can **represent a wide variety of interesting functions**
- First of its kind is proved by **George Cybenko**
  - in 1989
  - With the sigmoid activation functions



$$\begin{aligned}n_1(x) &= \text{Relu}(-5x - 7.7) \\n_2(x) &= \text{Relu}(-1.2x - 1.3) \\n_3(x) &= \text{Relu}(1.2x + 1) \\n_4(x) &= \text{Relu}(1.2x - .2) \\n_5(x) &= \text{Relu}(2x - 1.1) \\n_6(x) &= \text{Relu}(5x - 5)\end{aligned}$$

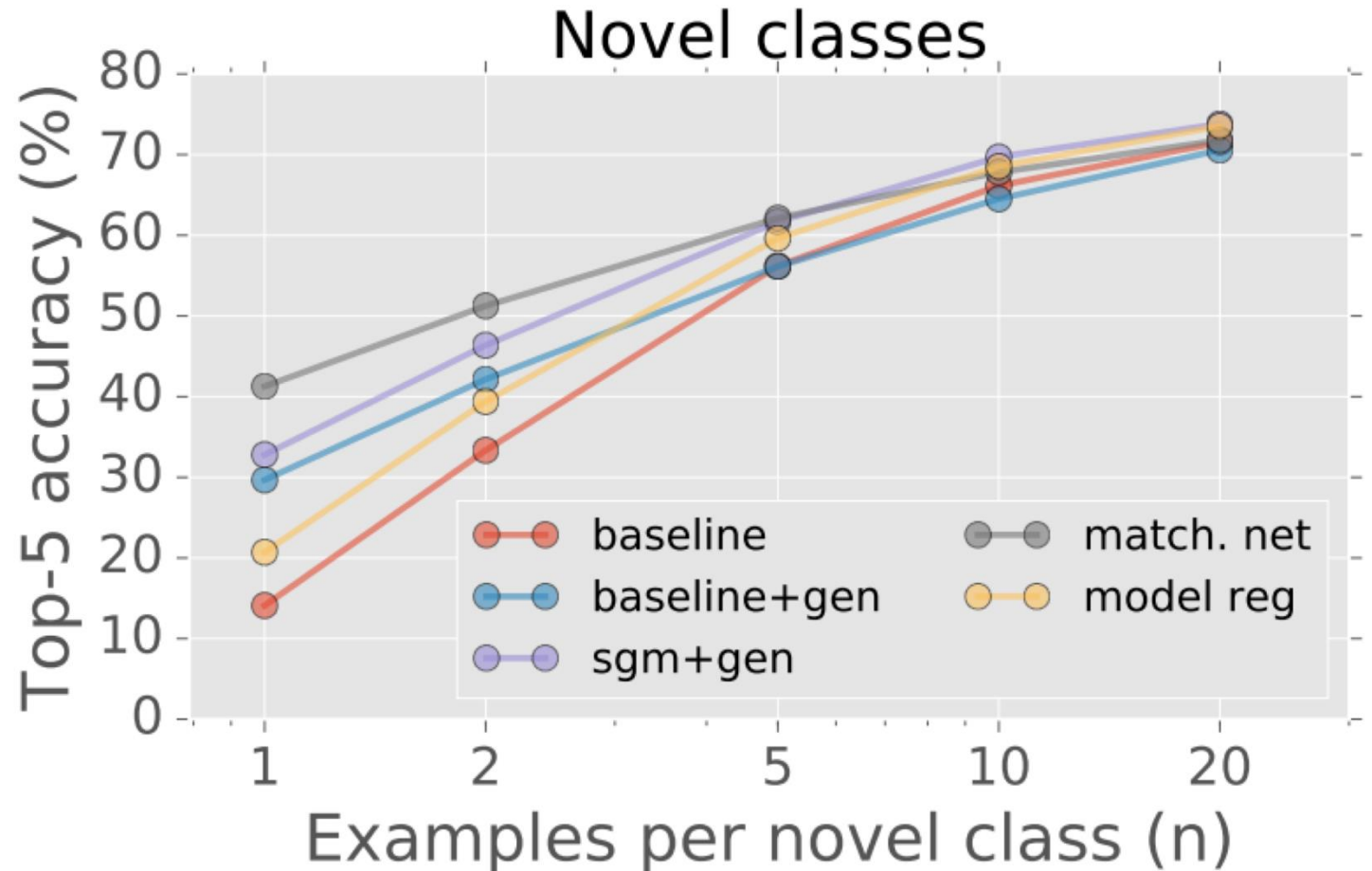
$$\begin{aligned}Z(x) &= -n_1(x) - n_2(x) - n_3(x) \\&\quad + n_4(x) + n_5(x) + n_6(x)\end{aligned}$$



# If we don't have large labeled data

In 611 classes in ImageNet Challenge dataset  
(1,000 classes in total)

Performance drops significantly



# Do we really need thousands of images?



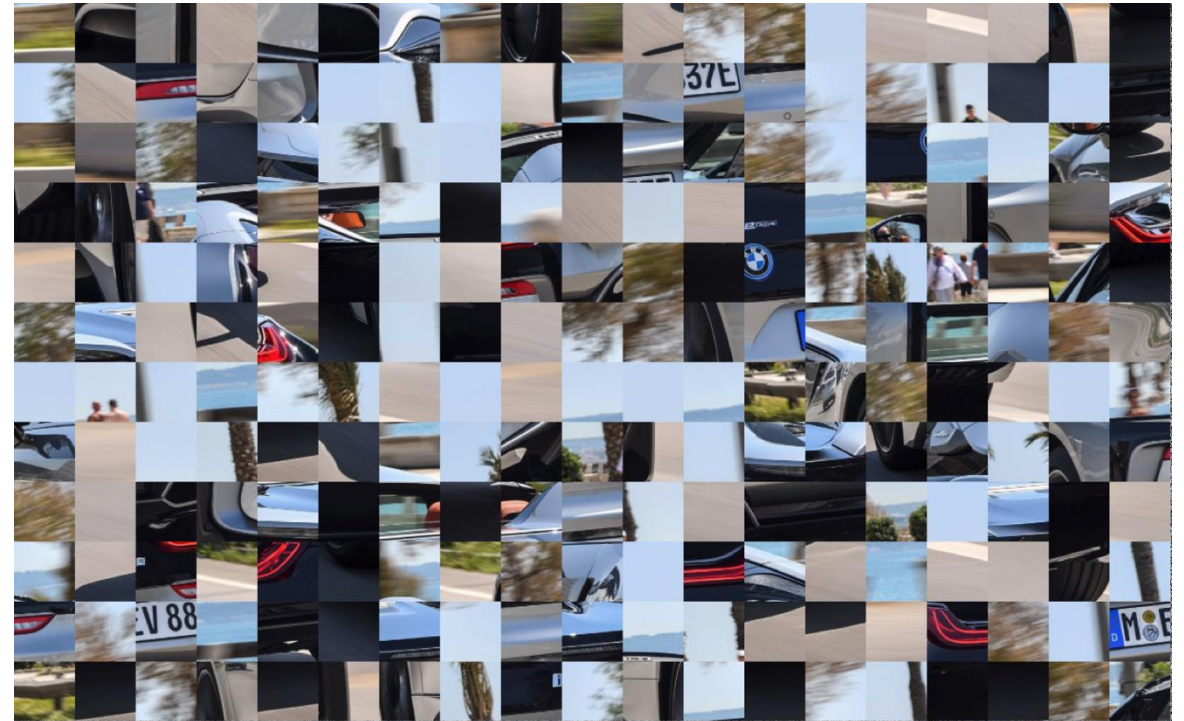
A flying car by AeroMobile

# Reason 1: Data encoding is not optimal

I am a boy who goes to a school

I am a boy who goes to a school

I am a boy who goes to a school



# So, we use CNN

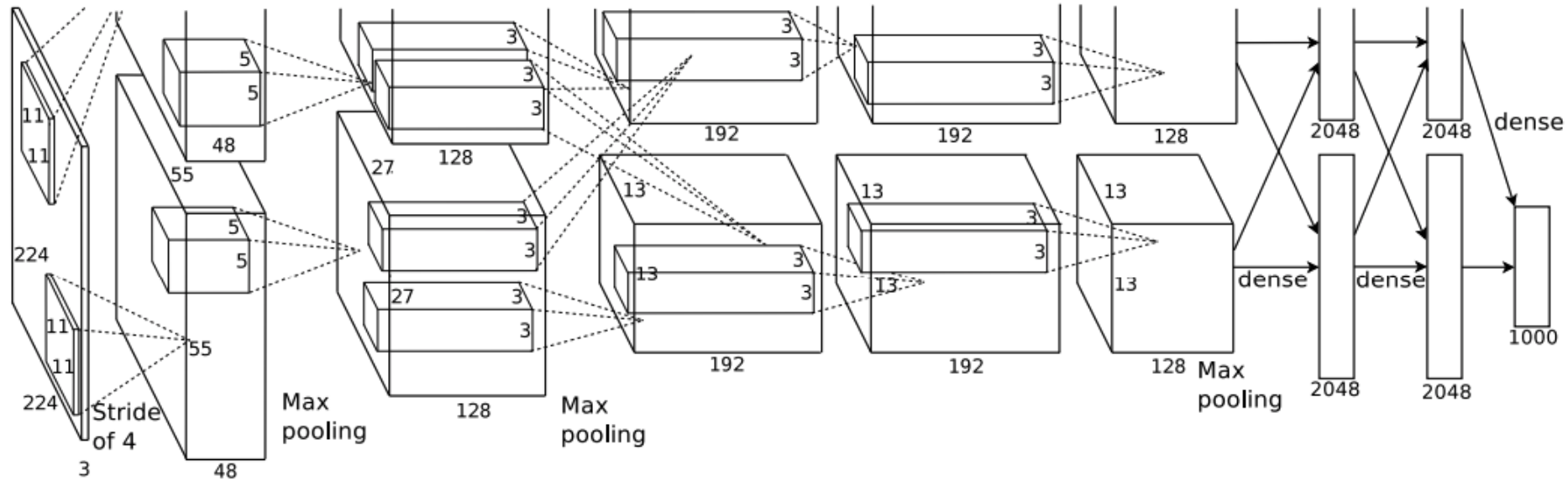


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.



# And we use transformers on the patches

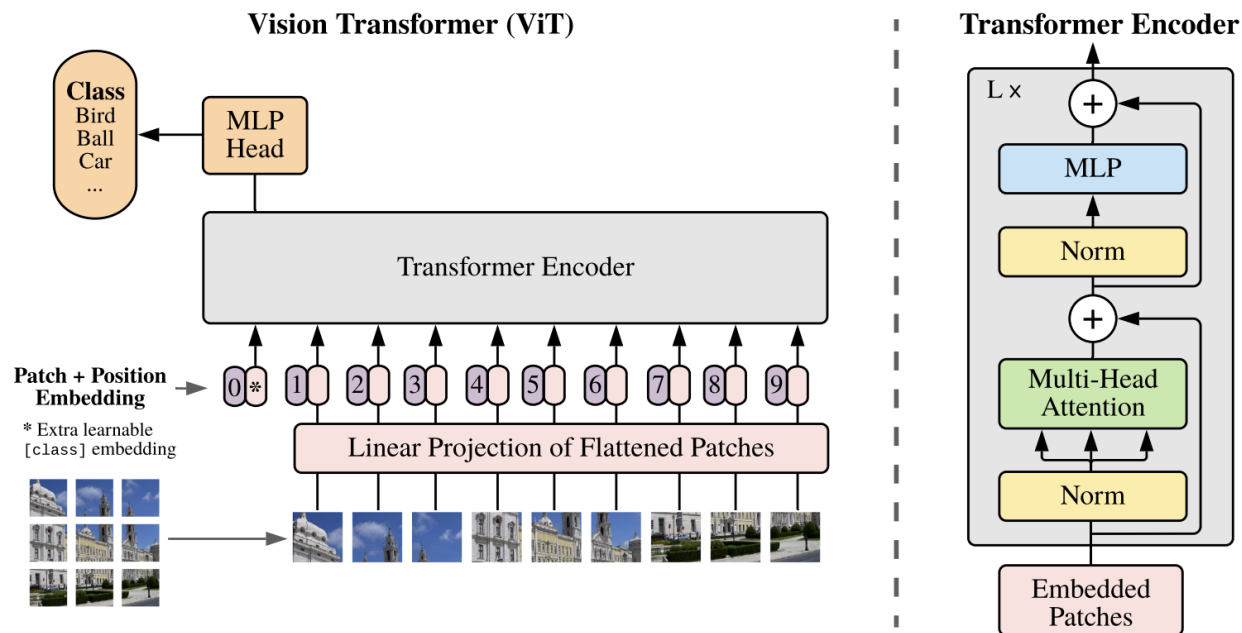


Figure 1: Model overview. We split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder. In order to perform classification, we use the standard approach of adding an extra learnable “classification token” to the sequence. The illustration of the Transformer encoder was inspired by Vaswani et al. (2017).



# But we still need quite large annotated data

The screenshot shows the ImageNet website interface. At the top, the 'IMAGENET' logo is on the left, a search bar with a 'SEARCH' button is in the center, and navigation links for 'Home', 'Explore', 'About', and 'Download' are on the right. Below the navigation bar, it states '14,197,122 images, 21841 synsets indexed'. A user status bar indicates 'Not logged in. Login | Signup'.

The main content area is for the 'Aircraft' category, described as 'A vehicle that can fly'. It shows statistics: '1510 pictures', '86.1 Popularity Percentile', and 'Wordnet IDs'. There are three tabs: 'Treemap Visualization', 'Images of the Synset' (which is active), and 'Downloads'.

On the left side of the 'Images of the Synset' tab, there is a hierarchical tree structure of synsets. The path highlighted is: ImageNet 2011 Fall Release (32326) → plant, flora, plant life (4486) → geological formation, formation (1112) → natural object (1112) → sport, athletics (176) → artifact, artefact (10504) → instrumentality, instrumentality (2760) → device (2760) → implement (726) → container (744) → hardware, ironware (0) → equipment (479) → ceramic (6) → means (0) → toiletry, toilet articles (57) → conveyance, transport (56) → vehicle (519) → bumper car, Dodgem craft (250) → aircraft (58) → boggy, bogie, b (58) → cruise missile (58) → heavier-than-air (58) → airplane, aerobically (58) → airliner (58) → airbus (58) → narrow (58) → wide (58) → amphibious (58) → biplane (58).

The main part of the 'Images of the Synset' tab displays a grid of 48 thumbnail images of various aircraft, including biplanes, propeller planes, jets, and military aircraft. At the bottom of the grid, there is a pagination bar with 'Prev', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '...', '43', '44', and 'Next'.

ImageNet, Deng et al., CVPR 2009

## Reason 2: One-hot supervision



Orange? Grapefruits?

|   |
|---|
| 0 |
| 0 |
| 1 |
| 0 |
| 0 |
| 0 |
| 0 |

But we use one-hot encoding and softmax

# Is one-hot supervision correct way of learning?

- A single label for an image (with multiple objects) should be noisy
  - Labeling is subjective



A Lady  
Sunglass  
Hand  
Lipstick



A Man  
Sunglass  
Mugshot  
Beard



A Boy  
Sunglass  
Bowtie  
Bubblegum



A Couple  
Sunglass  
Car  
Seat Belt



# Is one-hot supervision correct way of learning?

- A single label for an image (with multiple objects) should be noisy
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A Lady  
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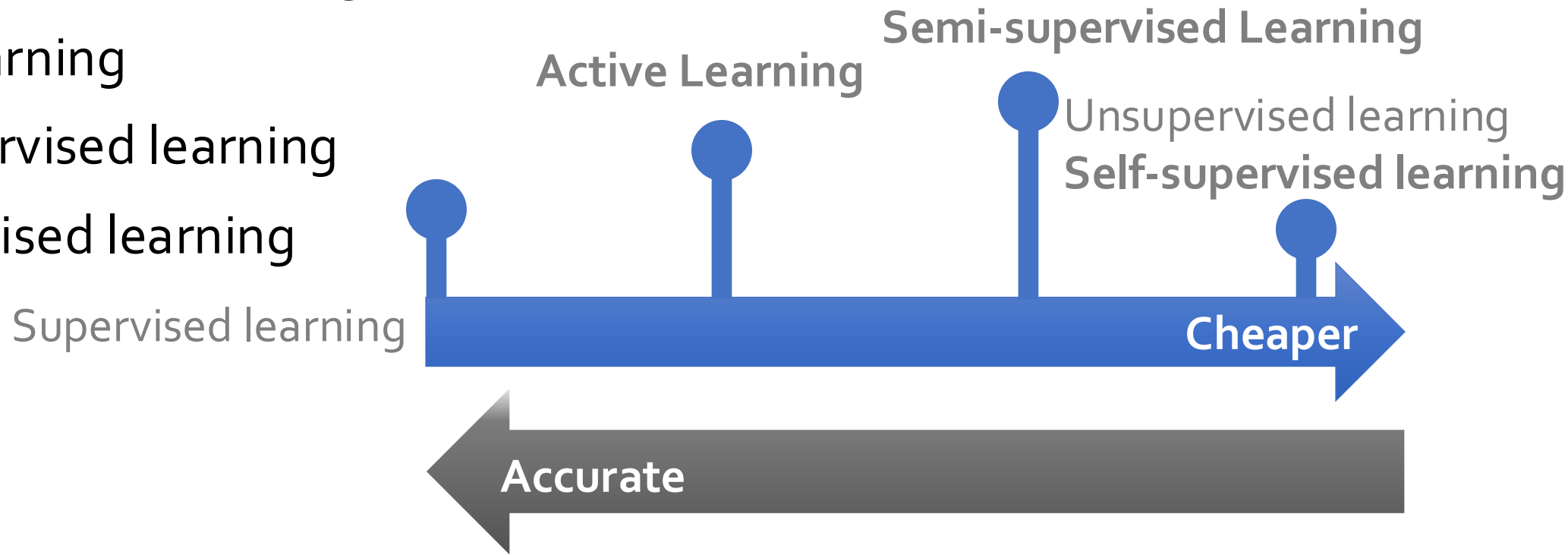
A Boy  
Sunglass  
Bowtie  
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A Couple  
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# Remedies in the literature

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# Few-shot, $N$ -shot, zero-shot learning

- **Few-shot ( $N$ -shot) learning:** Learning with a few ( $N$ ) examples for the relevant class
- **Zero-shot learning:** Learning with zero example for the relevant class

# Weakly supervised learning ideas

- Using inductive biases
  - Neural network architectures
  - Hand-crafted regularizers
  - Structure of data (inter sample relationship)
- Using unlabeled data
  - Semi-supervised learning
  - Self-supervised learning
  - Unsupervised learning
  - Using humans efficiently
    - Active learning
- Scaling models up

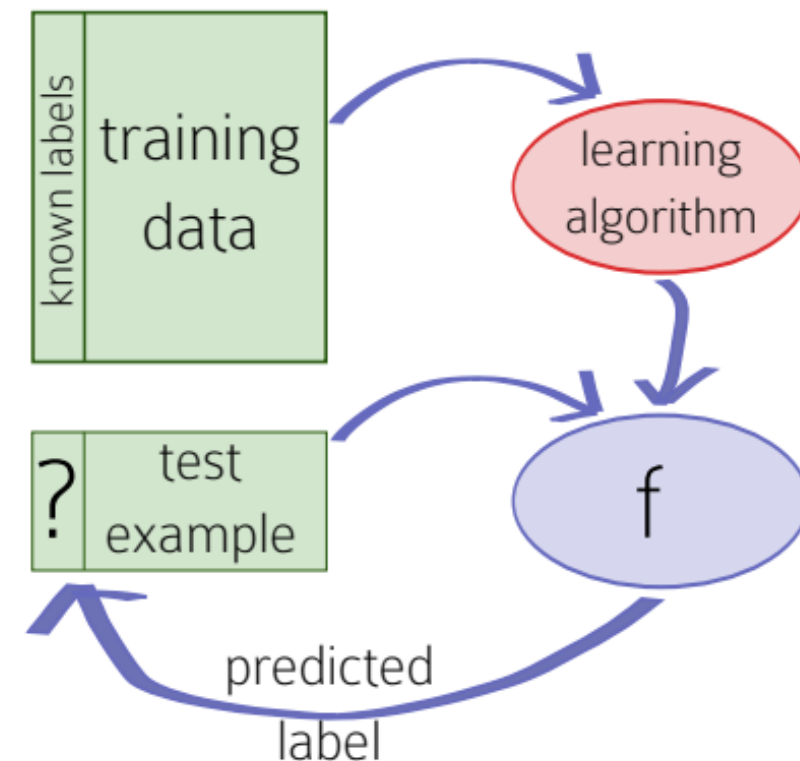
# What is inductive bias?

- Supervised learning = induction
- Predicting the future (test samples) based on the past (training samples)
- A bias to restrict the induction

**Inductive bias**



**regularizer**



# Typical learning objective

$$\min \text{ Loss } (L) + \text{ Regularizer } (\Omega)$$

- **Loss:** Things to be optimized (minimized)
  - Not much data  $\rightarrow$  less reliable
- **Regularizer:** An inductive bias
  - Rich inductive bias

# Examples of inductive bias in ML

- Maximum conditional independence
  - Used in Naïve bayes classifier
- Minimum cross-validation error
  - Choose hypothesis minimizing cross-validation error
- Maximum margin
  - Used in support vector machine (SVM)
- Minimum description length
  - Prefer shorter description. E.g., Occam's razor
  - Used in decision tree
- Minimum features
  - If some feature (attribute) is useless, delete it!
- Nearest neighbor
  - Assign a label by majority of its neighbors' labels
  - Used in K-nearest neighbor



# Examples of modern inductive bias

- Neural network architectures
- Hand-crafted regularizers
- Structure of data (inter sample relationship)

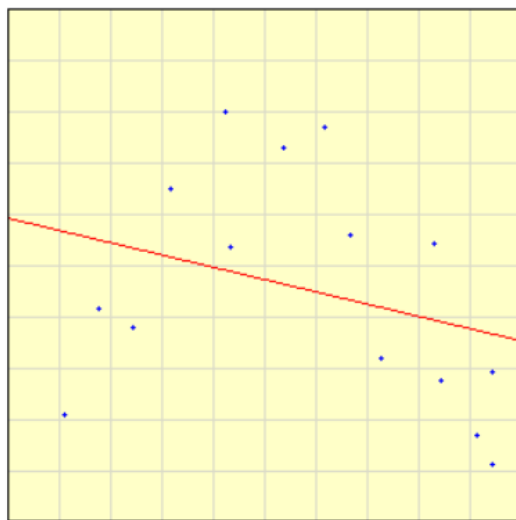
# Is much bias always good?

# No

- Need a good balance between *well fitting* and the *bias*
- Called **Bias-Variance Trade-off**

# Bias-Variance Tradeoff

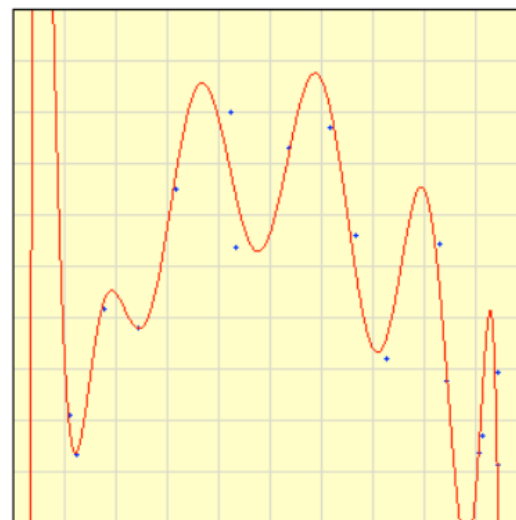
- Choice of hypothesis class introduces learning bias
  - More complex class  $\rightarrow$  less bias
  - More complex class  $\rightarrow$  more variance



Select points by clicking on the graph or press

Example

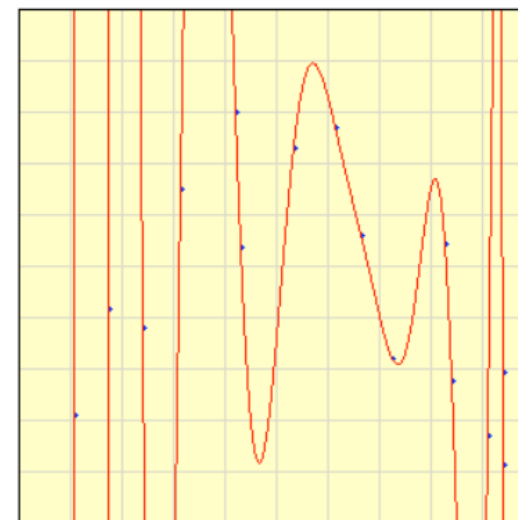
Degree of polynomial: 1  ☒ Fit Y to X  
☐ Fit X to Y



Select points by clicking on the graph or press

Example

Degree of polynomial: 13  ☒ Fit Y to X  
☐ Fit X to Y



Select points by clicking on the graph or press

Example

Degree of polynomial: 13  ☒ Fit Y to X  
☐ Fit X to Y

# Training error

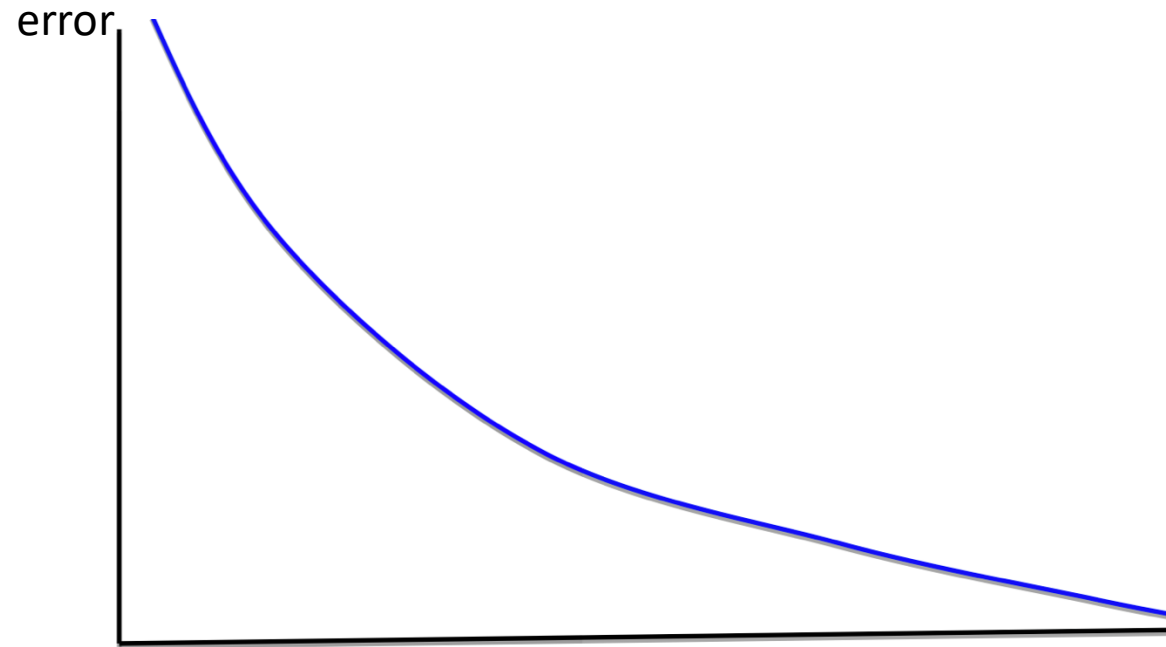
$$\mathbf{w}^* = \operatorname{argmin}_{\mathbf{w}} \sum_j (t(x_j) - \sum_i w_i h_i(x_j))^2$$

- Given a dataset (Training data)
- Choose a loss function
  - e.g., squared error (L2) for regression
- Training error: For a particular set of parameters, loss function on training data:

$$error_{train}(\mathbf{w}) = \frac{1}{N_{train}} \sum_{j=1}^{N_{train}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

# Training error as a function of model complexity

$$\text{error}_{\text{train}}(\mathbf{w}) = \frac{1}{N_{\text{train}}} \sum_{j=1}^{N_{\text{train}}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$



# True error

- Training set error can be poor measure of “quality” of solution
- **Prediction error (true error)**: We really care about error over all possibilities:

$$\begin{aligned} error_{true}(\mathbf{w}) &= E_{\mathbf{x}} \left[ \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 \right] \\ &= \int_{\mathbf{x}} \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 p(\mathbf{x}) d\mathbf{x} \end{aligned}$$



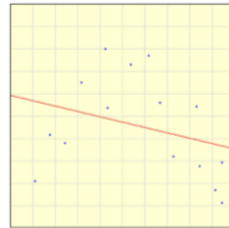
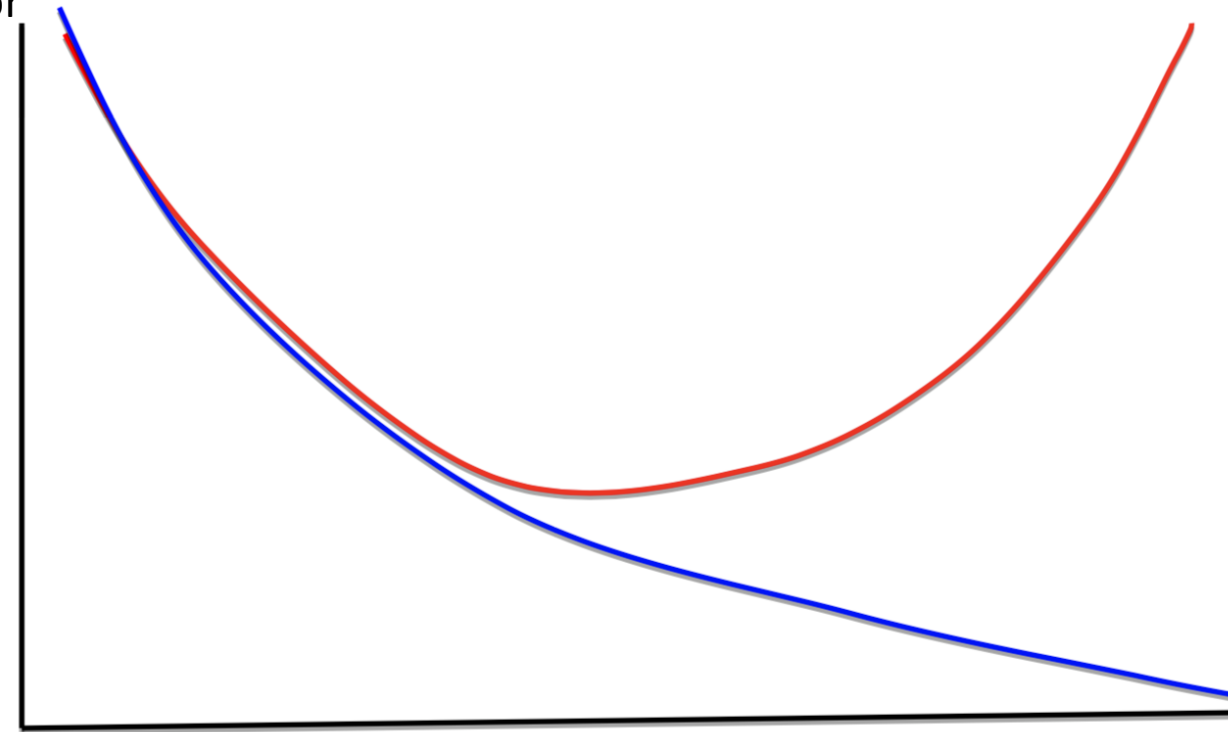


# True error as a function of model complexity

$$error_{train}(\mathbf{w}) = \frac{1}{N_{train}} \sum_{j=1}^{N_{train}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

$$error_{true}(\mathbf{w}) = \int_{\mathbf{x}} \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 p(\mathbf{x}) d\mathbf{x}$$

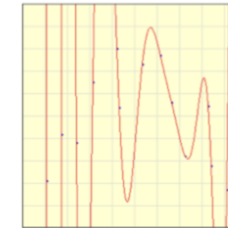
error



Select points by clicking on the graph or press [Example](#)

Degree of polynomial:  ☒ FXY to X ☐ FXY to Y

[Calculate](#) [View Polynomial](#) [Reset](#)



Select points by clicking on the graph or press [Example](#)

Degree of polynomial:  ☒ FXY to X ☐ FXY to Y

[Calculate](#) [View Polynomial](#) [Reset](#)

# Computing the true error

- To correctly predict error

- Hard integral!
- May not know  $t(x)$  for every  $x$ , may not know  $p(x)$

$$error_{true}(\mathbf{w}) = \int_{\mathbf{x}} \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 p(\mathbf{x}) d\mathbf{x}$$

- Monte Carlo integration (sampling approximation)

- Sample a set of i.i.d. points  $\{x_1, \dots, x_M\}$  from  $p(x)$
- Approximate integral with sample average

$$error_{true}(\mathbf{w}) \approx \frac{1}{M} \sum_{j=1}^M \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

# Why training set error doesn't approximate prediction error?

- Sampling approximation of prediction error:

**Because you are (optimistically) biased about computing error!**

- Training error is a good estimate for a single  $\mathbf{w}$ .
- But you optimized  $\mathbf{w}$  with respect to the **training error** and found  $\mathbf{w}$  that is good for this set of samples
- Training error is a (optimistically) biased estimate of the prediction error

Why is training set a bad measure of prediction error? :

# Prediction/Inference/Test set error

- Given a dataset, randomly split it into two parts
  - Training data:  $\{x_1, \dots, x_{N_{train}}\}$
  - Test data :  $\{x_1, \dots, x_{N_{test}}\}$
- Use training data to optimize parameters  $\mathbf{w}$
- Test set error: For the final solution  $\mathbf{w}^*$ , evaluate the error

$$error_{test}(\mathbf{w}) = \frac{1}{N_{test}} \sum_{j=1}^{N_{test}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

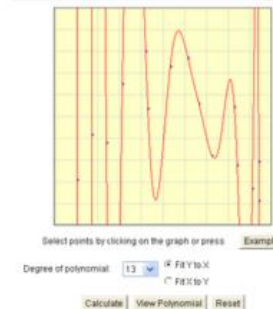
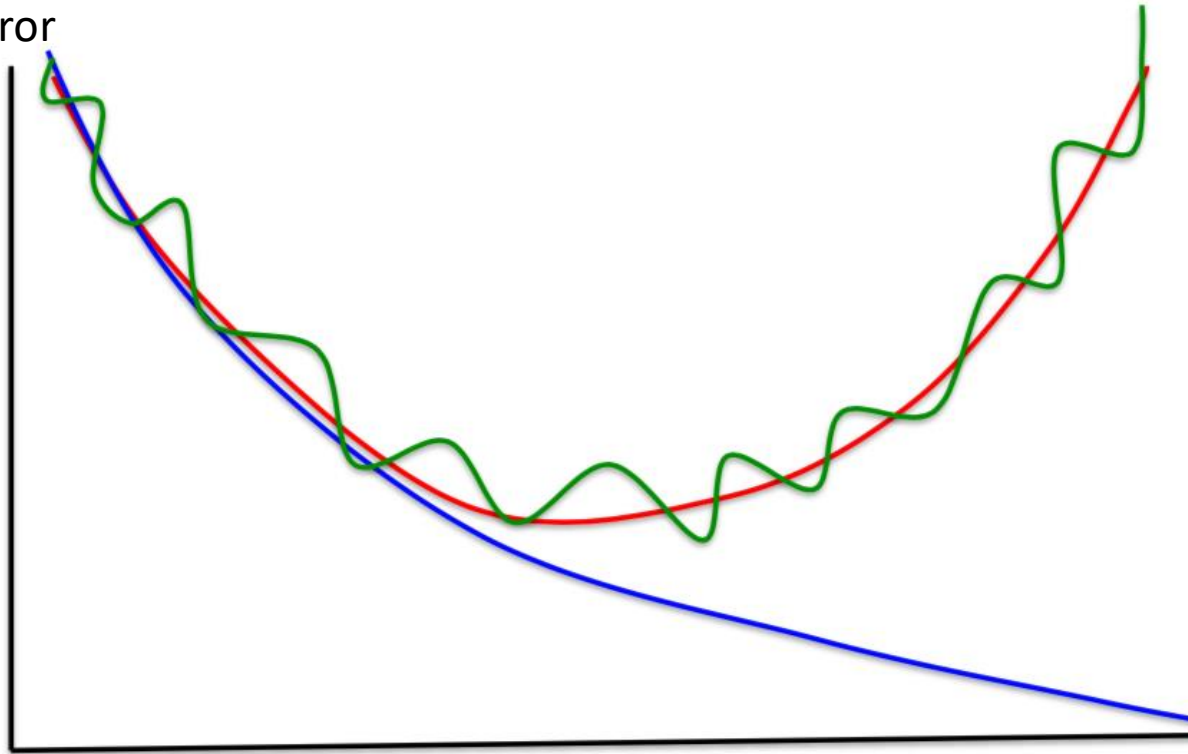
# Test set error as a function of model complexity

$$error_{train}(\mathbf{w}) = \frac{1}{N_{train}} \sum_{j=1}^{N_{train}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

$$error_{true}(\mathbf{w}) = \int_{\mathbf{x}} \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 p(\mathbf{x}) d\mathbf{x}$$

$$error_{test}(\mathbf{w}) = \frac{1}{N_{test}} \sum_{j=1}^{N_{test}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

error



# Overfitting

- Assume:
  - Data generated from a distribution  $D(X, Y)$
  - A hypothesis space  $H$
- Define: errors for hypothesis  $h \in H$ 
  - Training error:  $error_{train}(h)$
  - Data (true) error:  $error_{true}(h)$
- We say  $h$  overfits the training data if there exists an  $h' \in H$  such that
$$error_{train}(h) < error_{train}(h')$$

and

$$error_{true}(h) > error_{true}(h')$$

# Error estimators

- Ground (GND) truth / Gold standard:

$$error_{true}(\mathbf{w}) = \int_{\mathbf{x}} \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 p(\mathbf{x}) d\mathbf{x}$$

- **Training:** optimistically biased

$$error_{train}(\mathbf{w}) = \frac{1}{N_{train}} \sum_{j=1}^{N_{train}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

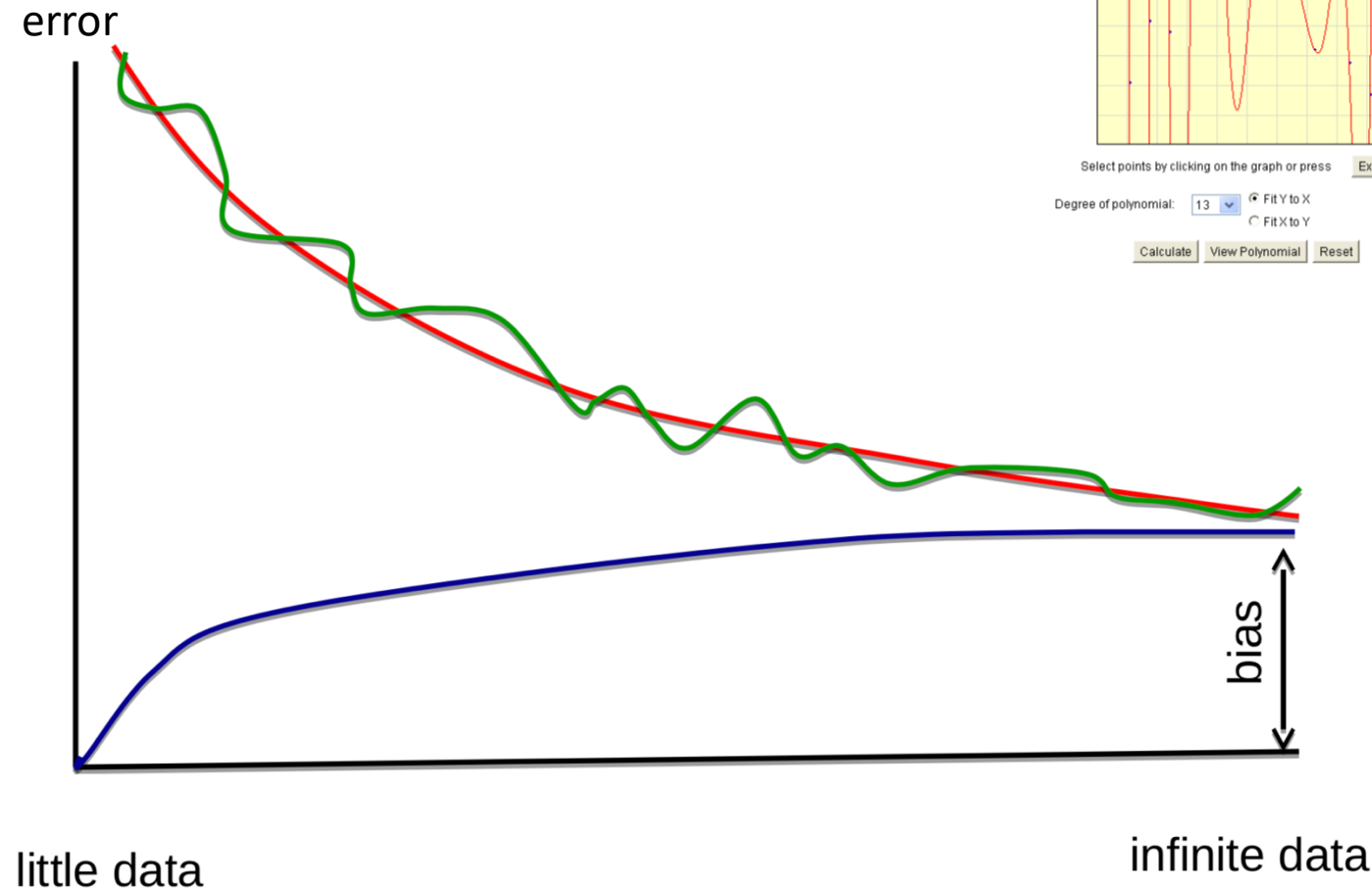
- **Test:** our final measure

$$error_{test}(\mathbf{w}) = \frac{1}{N_{test}} \sum_{j=1}^{N_{test}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$



# Error as a function of number of training examples for a fixed model complexity

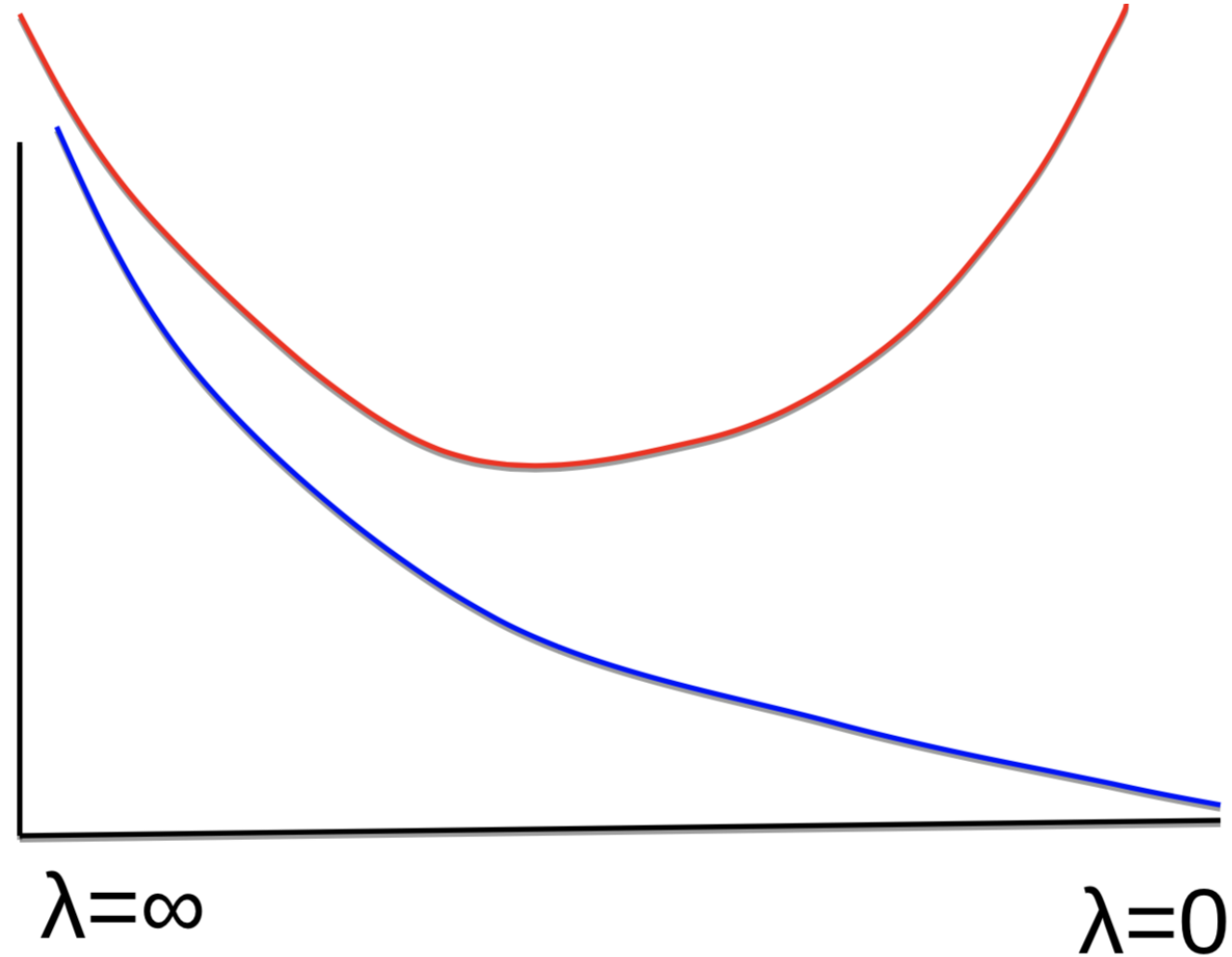
$$\begin{aligned} \text{error}_{\text{train}}(\mathbf{w}) &= \frac{1}{N_{\text{train}}} \sum_{j=1}^{N_{\text{train}}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2 \\ \text{error}_{\text{true}}(\mathbf{w}) &= \int_{\mathbf{x}} \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 p(\mathbf{x}) d\mathbf{x} \\ \text{error}_{\text{test}}(\mathbf{w}) &= \frac{1}{N_{\text{test}}} \sum_{j=1}^{N_{\text{test}}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2 \end{aligned}$$



# Error as a function of regularization parameter, fixed model complexity

$$error_{train}(\mathbf{w}) = \frac{1}{N_{train}} \sum_{j=1}^{N_{train}} \left( t(\mathbf{x}_j) - \sum_i w_i h_i(\mathbf{x}_j) \right)^2$$

$$error_{true}(\mathbf{w}) = \int_{\mathbf{x}} \left( t(\mathbf{x}) - \sum_i w_i h_i(\mathbf{x}) \right)^2 p(\mathbf{x}) d\mathbf{x}$$



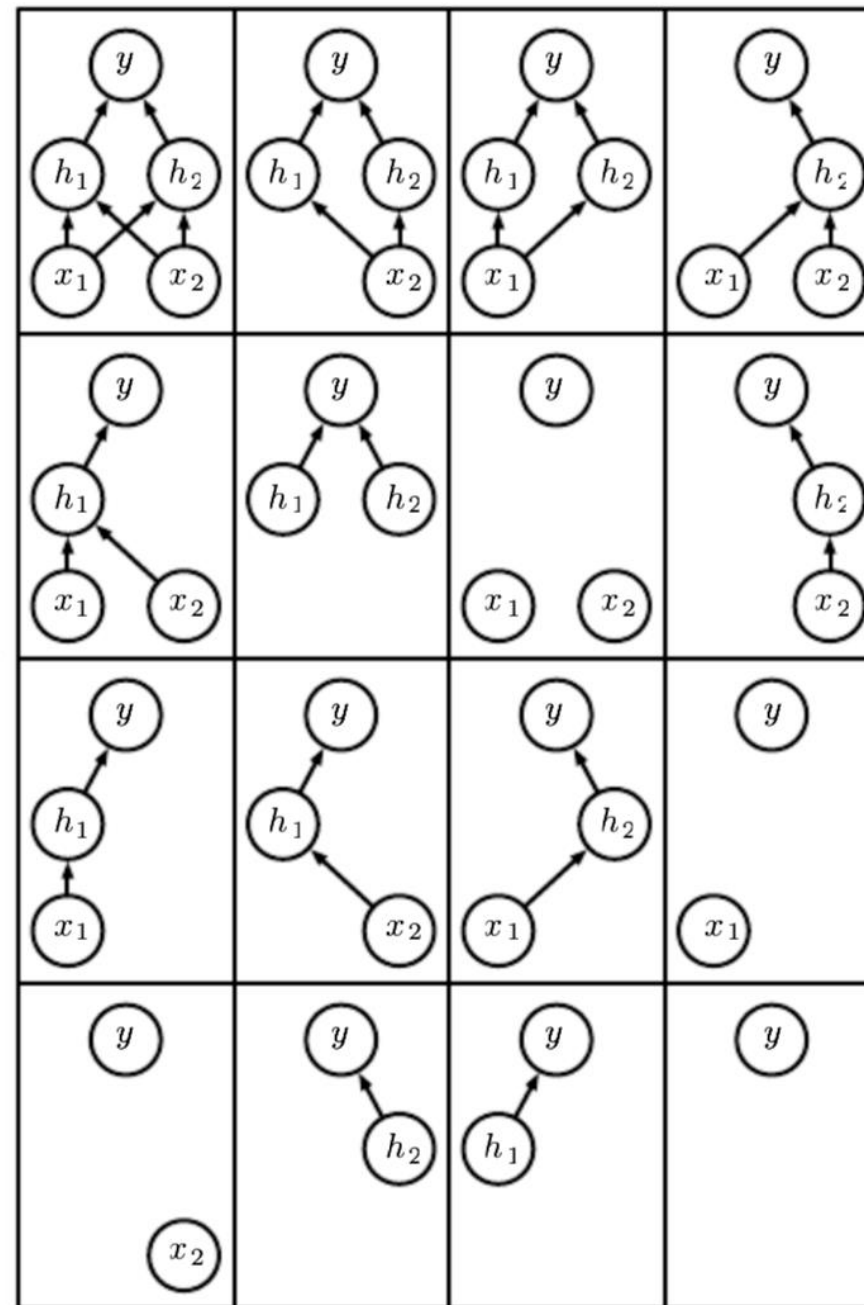
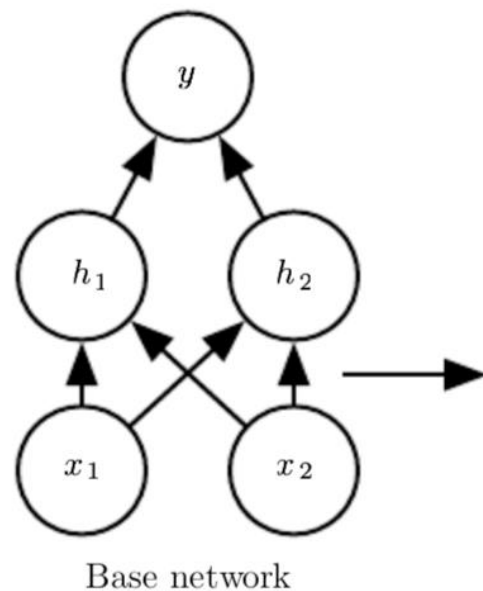
# Inductive bias 1 – Neural network architectures

- LeNet-5 (LeCun et al., 2000)
- AlexNet (Krizhevsky et al., 2012)
- VGG Net (Simonyan and Zisserman, 2015)
- ResNet (He et al., 2015)
- DenseNet (Huang et al., 2018)
- EfficientNet (Tan and Le, 2019)
- ViT (Vision Transformer) (Dosovitskiy et al., 2020)
- gMLP (Liu et al., 2021)
- ...

# Inductive bias 2 – Hand-crafted regularizers

- $L_2$
- $L_1$
- Dropout
- Data augmentation
- Early stopping

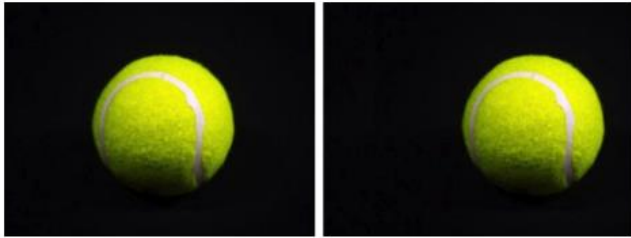
# DropOut and Ensemble



Ensemble of subnetworks

# Data Augmentation

- Reduce overfitting
- Robust to adversarial attack
- Transform your input to multiply the number of training samples



Translation



Original



Resized and  
horizontal flip



Rotation



Resized and  
crop

# Early Stopping

- Going far makes network converge to a local minimum
- Better to *less* converge to the local minimum
- Stop before the network completely converges

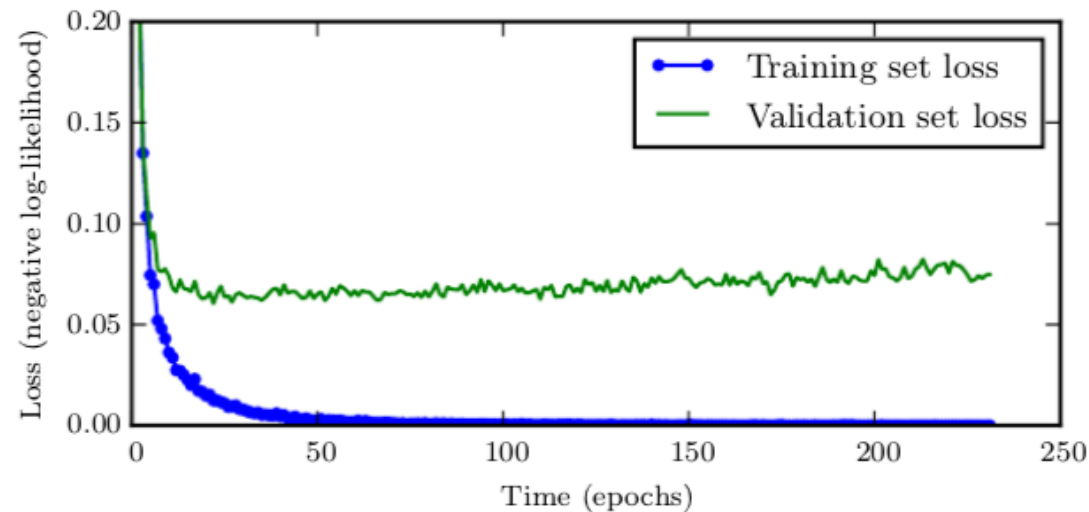


Figure 7.3: Learning curves showing how the negative log-likelihood loss changes over time (indicated as number of training iterations over the dataset, or **epochs**). In this example, we train a maxout network on MNIST. Observe that the training objective decreases consistently over time, but the validation set average loss eventually begins to increase again, forming an asymmetric U-shaped curve.



# Inductive bias 3 – Structure of data (inter/intra-sample relationships)

- Samples are dependent each other
  - They are similar and different in some senses
    - In different aspects
      - Visual attributes
      - Some semantics
    - Exclusive in labels
- Parts of samples are also related
  - Geometric relationship of parts is a unique feature

# Using unlabeled data

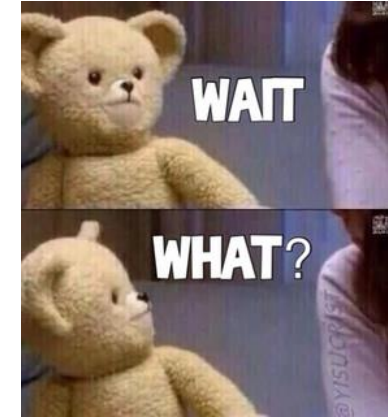
- Semi-supervised learning
  - Using a combined set of **labeled data** and **unlabeled data**
  - Usually
    - Inferring pseudo labels for the unlabeled data
    - Using the labeled and pseudo labeled unlabeled data
  - **Formal version:** Given  $D = \{\{X_i, y_i\}, \{X_j\}\}$ , find a function  $F_{pseudo}: \{X_j\} \rightarrow y_j$  to maximize the success of the supervised learning function  $F: \{\{X_i, Y_i\}, \{X_j, y_j\}\} \rightarrow Y$
- Active learning
  - (next page)



Pseudo-labeling the samples

# Using unlabeled data (cont'd)

- Active learning
  - Using a combined set of **labeled data** and **unlabeled data**
  - With a human labeler
  - **Goal:** to reduce the total amount of labeling efforts
  - **Formal version:** Given  $D = \{\{X_i, Y_i\}, \{X_j\}\}$ , reduce a number of labeling attempts of unlabeled portion of the dataset  $\{X_j\}$  (i.e., find true  $y_j$ ) while maximizing the success of the supervised learning function  $F: \{\{X_i, Y_i\}, \{X_j, y_j\}\} \rightarrow Y$

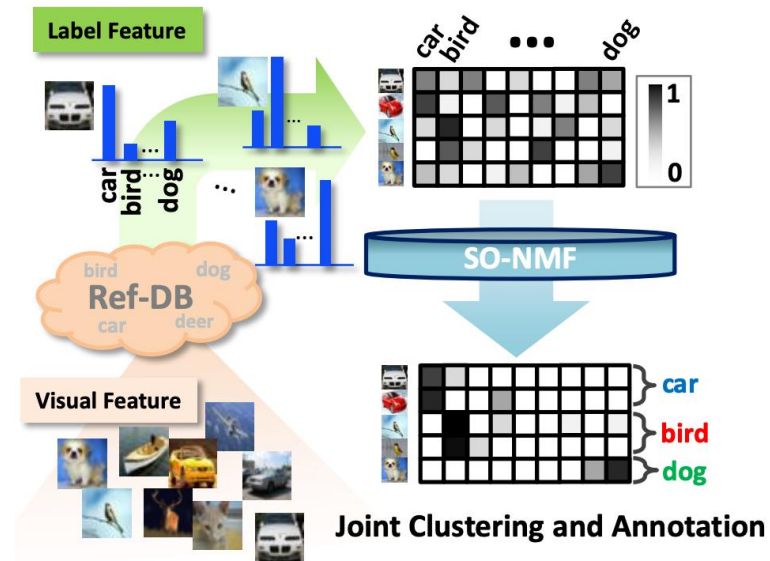


# Inductive bias 3 – Structure of data (inter/intra-sample relationships)

- In semi-supervised learning and active learning regimes
- Samples are dependent each other
  - They are similar and different in some senses
    - In different aspects
      - Visual attributes
      - Some semantics
    - Exclusive in labels
- Parts of samples are also related
  - Geometric relationship of parts is a unique feature

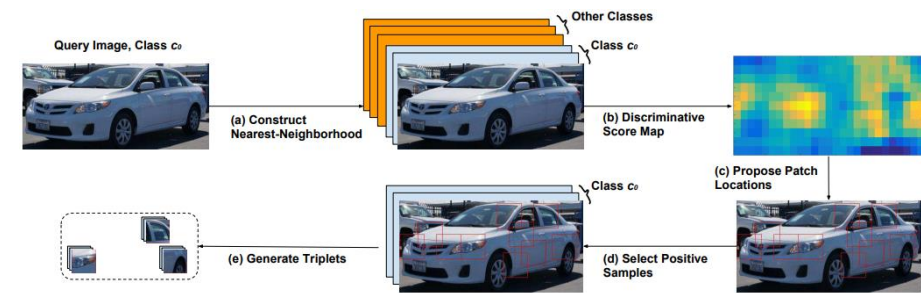
# Some examples of using structure of data

## Pseudo-labeling by the relationship



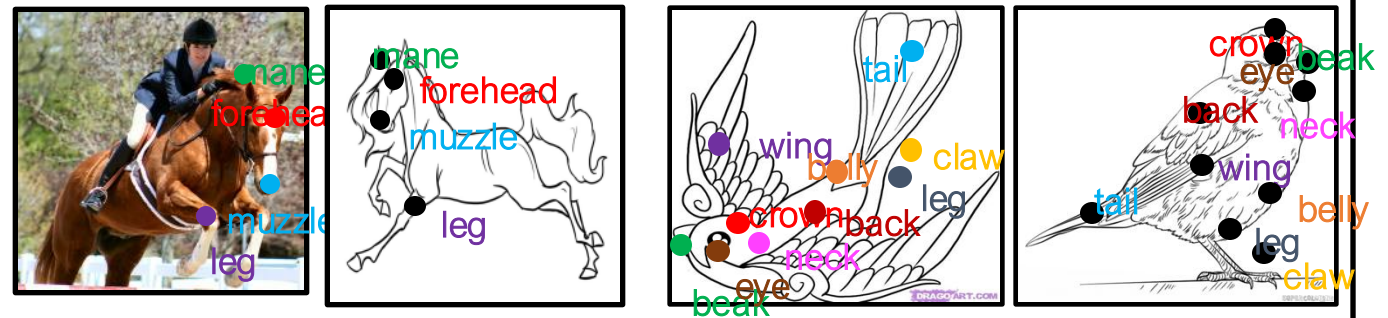
[Hong, Choi, Feyereisl, Han, Davis, *IEEE TPAMI* 2016]

## Fine-grained classification by part relationship



[Wang, Choi, Morariu, Davis, *CVPR* 2016]

## Cross-modal classification by part relationship



[Choi, Krishnamurthy, Kembhavi, Farhadi, *CVPR* 2018 *Spotlight*]

# Summary

- Few-shot learning
- Semi-supervised learning
- Active learning